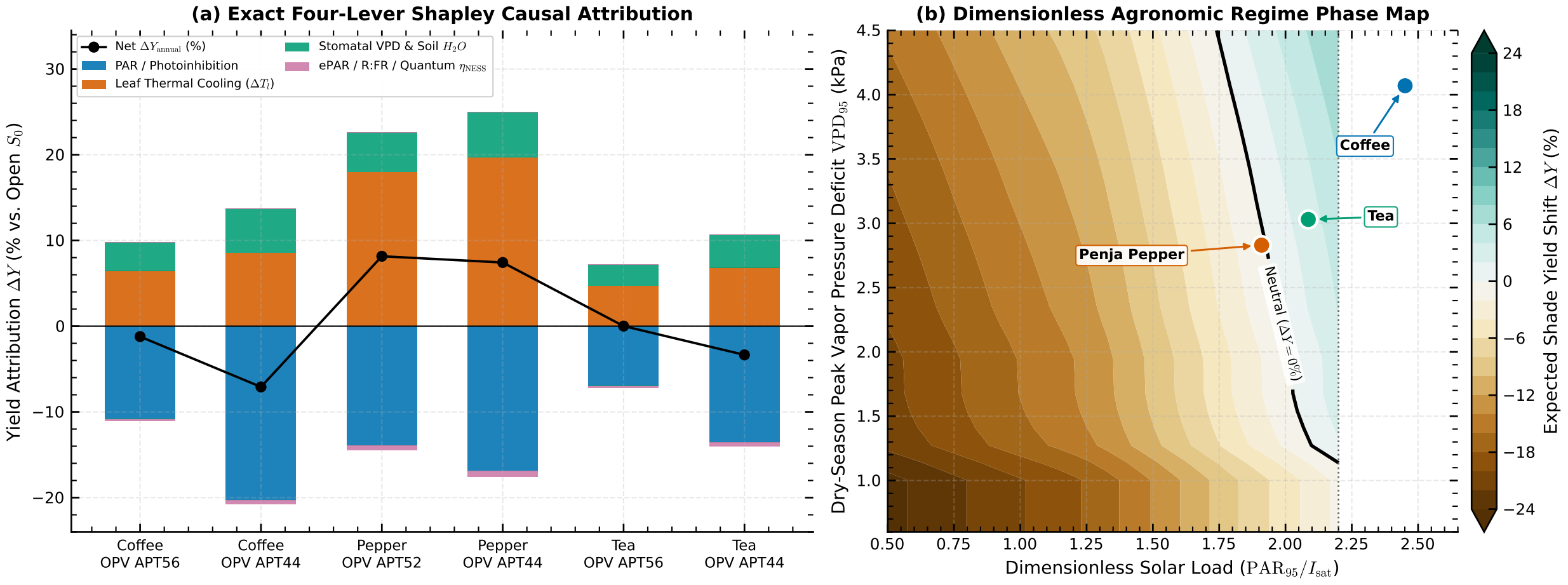

(a) Exact Four-Lever Shapley Causal Attribution
Net ΔY_{annual} (%)
PAR / Photoinhibition
Leaf Thermal Cooling (ΔT_l)
Stomatal VPD & Soil H_2O
ePAR / R:FR / Quantum η_{NESS}
Yield Attribution ΔY (% vs. Open S_0)
30
20
10
0
−10
−20
Coffee OPV APT56
Coffee OPV APT44
Pepper OPV APT52
Pepper OPV APT44
Tea OPV APT56
Tea OPV APT44
(b) Dimensionless Agronomic Regime Phase Map
Dry-Season Peak Vapor Pressure Deficit VPD_{95} (kPa)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25
2.50
Dimensionless Solar Load ($\text{PAR}_{95}/I_{\text{sat}}$)
Coffee
Tea
Penja Pepper
Neutral ($\Delta Y = 0$ %)
Expected Shade Yield Shift ΔY (%)
24
18
12
6
0
−6
−12
−18
−24